

MODEL VALIDATION, OVERFITTING CONTROL & HYPERPARAMETER TUNING

Artificial Intelligence & Machine Learning – Task 3

Submitted By

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1. Introduction

Machine Learning models are expected not only to perform well on training data but also to generalize effectively to unseen data. One of the major challenges in machine learning is overfitting, where a model learns patterns specific to the training data and fails to perform reliably on new data.

This project focuses on model validation, overfitting detection, cross-validation, and hyperparameter tuning techniques. The California Housing Dataset was used to build and evaluate regression models. The objective was to improve model reliability and predictive performance through systematic validation and optimization.

2. Objectives

The objectives of this project were:

- Detect overfitting in machine learning models.
 - Apply Cross-Validation for reliable model evaluation.
 - Perform Hyperparameter Tuning using GridSearchCV.
 - Compare baseline and optimized models.
 - Select the best-performing model based on evaluation metrics.
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3. Dataset Description

The California Housing Dataset available in Scikit-Learn was used for this project.

Features

- Median Income (MedInc)
- House Age (HouseAge)
- Average Rooms (AveRooms)
- Average Bedrooms (AveBedrms)
- Population
- Average Occupancy (AveOccup)

- Latitude
- Longitude

Target Variable

- Median House Value

The dataset is widely used for regression tasks and contains housing information collected from districts across California.

4. Methodology

4.1 Data Preprocessing

The following preprocessing steps were performed:

1. Dataset loading using Scikit-Learn.
2. Feature-target separation.
3. Feature scaling using StandardScaler.
4. Train-test split using an 80:20 ratio.

Feature scaling was applied to ensure uniform treatment of all features during model training.

4.2 Baseline Model Development

Three machine learning models were initially trained and evaluated:

Linear Regression

Linear Regression was used as the baseline model to establish a performance benchmark.

Ridge Regression

Ridge Regression was applied to introduce regularization and reduce the risk of overfitting.

Decision Tree Regressor

Decision Tree Regression was used to capture non-linear relationships present in the housing dataset.

5. Overfitting Analysis

Overfitting occurs when a model learns training data excessively and performs poorly on unseen data.

To detect overfitting, the Decision Tree model's training and testing performances were compared using RMSE.

Observation

The Decision Tree model achieved improved predictive performance compared to linear models. However, unrestricted tree growth can lead to overfitting and poor generalization.

Therefore, model validation and hyperparameter tuning were performed to improve reliability.

6. Cross-Validation

A 5-Fold Cross Validation strategy was implemented to obtain a reliable estimate of model performance.

Cross Validation RMSE Scores

- 0.8877
- 0.8278
- 0.8985
- 0.9451
- 0.9195

Average Cross Validation RMSE

0.8957

Importance of Cross Validation

- Reduces dependency on a single train-test split.
- Provides a realistic estimate of model performance.
- Improves confidence in model selection.
- Helps identify unstable models.

Cross-validation confirmed that the model performance remained reasonably consistent across multiple data partitions.

7. Hyperparameter Tuning Using GridSearchCV

Hyperparameters significantly affect model behavior and predictive performance.

GridSearchCV was used to identify the best combination of Decision Tree parameters.

Hyperparameters Tuned

- max_depth
- min_samples_split

Process

1. Multiple parameter combinations were generated.
2. Cross-validation was applied to each combination.

3. Performance was evaluated using RMSE.
4. The best-performing parameter set was selected automatically.

This process helped reduce overfitting and improve model generalization.

8. Evaluation Metrics

The following metrics were used:

Root Mean Squared Error (RMSE)

RMSE measures the average prediction error between actual and predicted values.

Lower RMSE indicates better model performance.

R² Score

R² Score measures the proportion of variance explained by the model.

Higher R² values indicate stronger predictive capability.

9. Results and Discussion

Model Performance Comparison

Model	RMSE	R ² Score
Linear Regression	0.7456	0.5758
Ridge Regression	0.7456	0.5758
Decision Tree	0.7030	0.6228
Optimized Decision Tree	0.6454	0.6821

Discussion

The Linear Regression and Ridge Regression models achieved nearly identical performance, indicating that regularization had minimal influence on this dataset.

The Decision Tree model improved performance over both linear models by capturing non-linear relationships within the data.

However, Decision Trees are susceptible to overfitting. Therefore, GridSearchCV was applied to optimize model complexity.

After hyperparameter tuning:

- RMSE improved from **0.7030** to **0.6454**.
- R² Score improved from **0.6228** to **0.6821**.

These improvements demonstrate that the optimized model generalized better and produced more accurate predictions.

10. Final Model Selection

The Optimized Decision Tree Regressor was selected as the final model because it achieved:

Best RMSE

0.6454

Best R² Score

0.6821

Advantages

- Lowest prediction error among all models.
- Highest explanatory power.
- Better generalization performance.
- Reduced overfitting through hyperparameter tuning.
- Reliable evaluation through Cross Validation.

The optimized Decision Tree was therefore considered the most suitable model for deployment.

11. Technologies Used

- Python
 - Pandas
 - NumPy
 - Matplotlib
 - Scikit-Learn
 - Jupyter Notebook
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12. Learning Outcomes

Through this project, the following concepts were explored:

- Overfitting and Underfitting
- Model Validation Techniques
- Cross Validation
- Hyperparameter Tuning
- GridSearchCV
- Bias-Variance Tradeoff

- Regression Evaluation Metrics
- Model Selection and Optimization

These concepts are essential for developing reliable and production-ready machine learning systems.

13. Conclusion

This project demonstrated the importance of model validation and optimization in machine learning workflows.

Cross-validation provided a reliable estimate of model performance across multiple data partitions, while GridSearchCV successfully identified improved hyperparameter settings for the Decision Tree model.

The optimized Decision Tree achieved the lowest RMSE (0.6454) and highest R^2 Score (0.6821), making it the best-performing model in this study.

Overall, the project successfully improved model reliability, reduced overfitting, and enhanced predictive performance through systematic validation and tuning techniques.